Market Power and Contagion in Interbank Networks

An Agent-Based Model of Systemic Risk

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Outline

Abstract: ABM, Interbank Market, and Contagion

- We use **Agent-Based Models** to study systemic risk in interbank credit networks.
- The dynamics are driven by two key parameters: the **probability of attachment** p of Erdős–Rényi graphs as interbank connections generator (exogenous) and the **interest rate** r (endogenous).
- As p increases, the network becomes more connected, forming a giant component. Interest rates decline due to greater competition and reduced market power.
- However, higher connectivity also raises the probability of borrower bankruptcy. Excessively low interest rates, maintained for too long, can be **detrimental to the stability of the system.**

Outline

The 2008 Crisis: Background

- A prolonged period of low interest rates encouraged excessive risk-taking and increased bank leverage.
- Standard competitive equilibrium models failed to capture the complex interactions that amplify crises.
- The interbank market became a **contagion vector**: failure of one bank triggered a domino effect through interlocking exposures, revealing how interconnectedness can magnify systemic fragility.

The Risk-Sharing Trade-off

Two opposing effects of network connectivity:

- **Benefit:** Increasing connectivity allows **risk sharing**—banks diversify exposure across more counterparties, lowering interest rates.
- **Cost:** Excessive connectivity increases **systemic risk**—default propagates through the network, and lenders serving a larger number of clients face higher bankruptcy exposure.
- Evidence: a **non-monotonic relationship** exists—low connectivity stabilizes by diluting shocks, but high connectivity facilitates cascading failures. Taken together, these effects imply that excessively low interest rates increase both bankruptcies and contagion.

Channels of Contagion in Financial Crises

Three primary propagation mechanisms of systemic failure:

1. **Bank runs:** self-fulfilling panics where depositors withdraw mass funds simultaneously.
2. **Asset price contagion:** fire-sale spirals that depreciate assets across the system.
3. **Interlocking exposures:** direct linkages through the interbank market—default of one bank triggers defaults in connected banks.

The model distinguishes three failure channels: **rationing** (borrower cannot secure sufficient funding), **contagion** (lender suffers losses from borrower default), and **repayment failure.** At low connectivity, rationing dominates; at high connectivity, repayment failure becomes the primary risk.

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Model Overview

- ABM with N banks in a sequential economy $t = \{0, 1, \dots, T\}$.
- Banks connected through an **Erdős–Rényi** random graph $G(N, p)$: each edge exists with exogenous probability p (probability of attachment).
- Three interacting sectors: goods market, credit market, interbank market.
- The **interest rate** r is the key endogenous variable determining loan relationships between borrowers and lenders.
- Goal: analyze how **connectivity** (p) and **market power** affect system stability through the endogenous adjustment of r.

Balance Sheet

Accounting Identity

$$L_i^t + C_i^t + R_i^t = D_i^t + E_i^t$$

- **Assets:** Long-term loans (L), liquidity (C), reserves (R).
- **Liabilities:** Deposits (D), equity (E).
- Reserves set by regulation: $R_i^t = \hat{r} D_i^t$ with $\hat{r} = 0.02$.

Liquidity Shocks

Deposits are exogenously shocked each period:

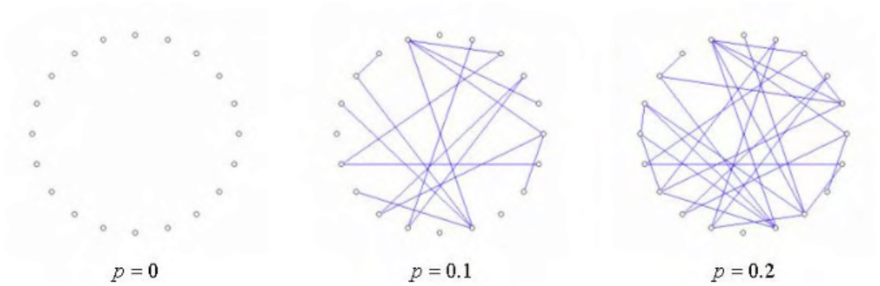
$$D_i^t = D_i^{t-1}[\mu + \omega \cdot U(0,1)]$$

- μ: expected growth rate (0.7).
- ω: shock magnitude (0.35).
- **Potential lenders:** banks with positive net deposits (after reserves).
- **Potential borrowers:** banks with negative net deposits exceeding liquidity.

Network Structure

- Bilateral credit lines formed at the start of each period.
- Connectivity controlled by parameter p in the Erdős–Rényi graph.
- Each borrower is matched to **exactly one** lender per period.
- The **Giant Component Size (GCS)** measures network integration.

Erdős–Rényi Random Graph



Each of the $\binom{N}{2}$ possible edges is included independently with probability p. For $N = 50$ banks:

- $p = 0$: isolated banks, no interbank market.
- $p = 1/N = 0.02$: **percolation threshold**—giant component emerges.
- $p > 0.05$: network is saturated ($> 92\%$ of banks connected).

Network Structure

The Erdős–Rényi model does not attempt to replicate the entire interbank market; rather, it illustrates how connection density alters the probability and scale of contagion.

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Expected Profit

Banks are risk-neutral in perfect competition. Expected profit from lending to borrower j:

$$E[\Pi_{i,j}^t] = p_j^t(r_{i,j}^t - m)c_{i,j}^t - (1 - p_j^t)m \cdot c_{i,j}^t$$

- m: screening cost (0.015).
- p_j^t: probability of borrower survival.
- $r_{i,j}^t$: interest rate; $c_{i,j}^t$: lending capacity.
- The interest rate is the key variable determining the profitability and risk of each loan relationship.

Interest Rate Derivation

Setting $E[\Pi] = 0$ (zero-profit condition):

$$r_{i,j}^t = \frac{m}{p_j^t}$$

Introducing **market power** $\psi \in [0, 1)$:

$$\boxed{r_{i,j}^t = \frac{m}{p_j^t(1 - \psi)}}$$

- $\psi = 0$: competitive (actuarially fair) rate.
- $\psi > 0$: monopolistic markup from information asymmetry.
- As connectivity p increases, competition intensifies, market power ψ declines, and the equilibrium interest rate falls.

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Endogenous Market Power

Market power evolves endogenously with bank size:

$$\psi_i^t = \frac{E_i^t}{E_i^{t,\max}}$$

- Larger banks (higher equity) exercise more market power.
- Borrower default probability: $p_{b,j}^t = E_j^t / E_j^{t,\max}$.
- Leverage: $\lambda_j^t = d_j^t / E_j^t$.
- As p increases, the market becomes more competitive — borrowers can choose among more lenders, reducing lenders' market power. However, this also allows lenders to serve more clients, increasing their exposure to borrower default.

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Equity Evolution

$$E_i^t = E_i^{t-1} + \sum_j l_{i,j}^t r_{i,j}^t - \sum_{j \subseteq \theta_i^t} \left[B_{i,j}^t - \tilde{L}_j^t \right]$$

- **Bankruptcy:** bank exits if $E_i^t < 0$.
- **Fire sales:** last-resort option (blocked in this model to amplify contagion effects).
- Failed banks replaced by new entrants with small initial capital.
- Endogenous interest rates affect bank earnings; low rates compress margins but also reduce the cost of borrowing, generating opposing effects on stability.

Failure Channels

Three distinct causes of bank failure:

1. **Rationing:** borrower cannot secure sufficient funding.
2. **Contagion:** lender suffers losses from borrower default (bad debt).
3. **Repayment failure:** fully-funded borrower cannot repay loan or interest.

Rationing dominates at low connectivity (fewer lending opportunities). **Repayment failure dominates** at high connectivity, as lenders serve more clients and the probability of borrower bankruptcy rises with lower interest rates.

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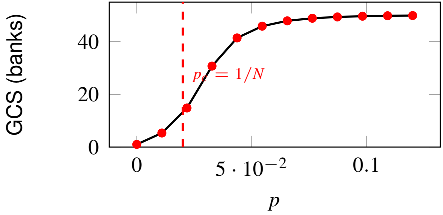
Phase Transition

- **Percolation threshold:** GCS jumps from ~ 1 to ~ 46 banks (92%) as p crosses the critical value $1/N$.
- **Low p:** rationing-dominated, high interest rates ($r \approx 30$), few loans granted.
- **High p:** repayment-failure-dominated, low interest rates ($r \approx 1.8$), saturated network.
- As the market becomes more competitive, lenders serve more clients, the probability of bankruptcy rises, and bad debt accumulates.

Phase Transition



Percolation threshold for

Erdős–Rényi graph:

$$p_c = \frac{1}{N} = \frac{1}{50} = 0.02$$

p	GCS
0.00001	1
0.011	5
0.022	**15**
0.033	31
0.044	41
0.055	**46**
0.065	48
0.087	49
≥ 0.11	50

Connectivity and Systemic Risk

- As p increases, the interest rate drops from 30 (upper bound) to ~ 1.65, reflecting reduced market power.
- Bankruptcies remain stable at ~ 7–7.5, with the composition shifting from rationing to repayment failure.
- Active loans saturate at ~ 13 (26% of banks), indicating that lenders can serve more clients.
- Bad debt grows rapidly with connectivity (from 0 to ~ 1.3), as more loans translate into higher default exposure.
- Taken together, these results show that an interest rate that remains too low for too long can be **detrimental to the stability of the system**.

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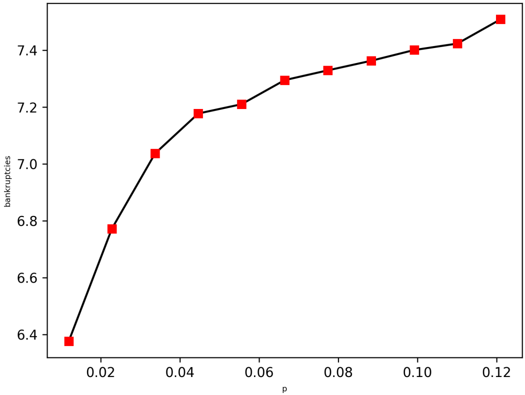
Bankruptcies



p	bankruptcies
0.012	6.3762
0.023	6.7724
0.034	7.0369
0.045	7.1774
0.056	7.2100
0.066	7.2948
0.077	7.3292
0.088	7.3628
0.099	7.4007
0.110	7.4234
0.121	7.5088

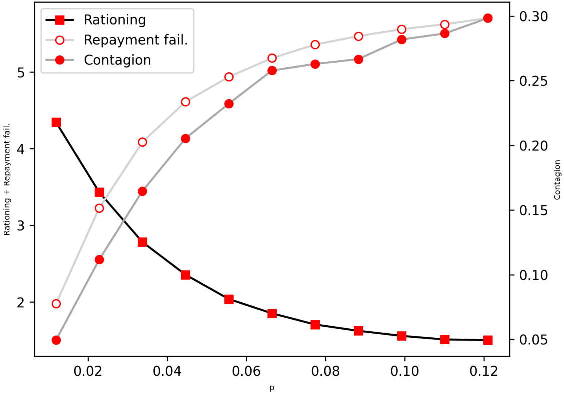
Monte Carlo simulations with $\mathcal{MC} = 50$ runs. $N = 50$ banks, $T = 1000$ periods, and p varying from 0 to 0.12 in increments of 0.01.

Initial conditions for each bank:

Assets	**Liabilities**
$C_0^i = 5 - 0.18 = 4.82$	$D_0^i = 9$
$L_0^i = 5$	$E_0^i = 1$
$R_0^i = 0.18$	

Bankruptcies decomposition

As p increases, the channel of failure shifts from rationing (few lenders available) to repayment failure (low rates, more clients served). Monte Carlo simulations with $\mathcal{MC} = 50$ runs. $N = 50$ banks, $T = 1000$ periods, and p varying from 0 to 0.12 in increments of 0.01.



p	Rat.	Cont.	Fail.
0.012	4.35	0.05	1.98
0.023	3.43	0.11	3.23
0.034	2.78	0.16	4.09
0.045	2.36	0.21	4.61
0.056	2.04	0.23	4.94
0.066	1.85	0.26	5.18
0.077	1.71	0.26	5.36
0.088	1.63	0.27	5.47
0.099	1.56	0.28	5.56
0.110	1.51	0.29	5.62
0.121	1.51	0.30	5.71

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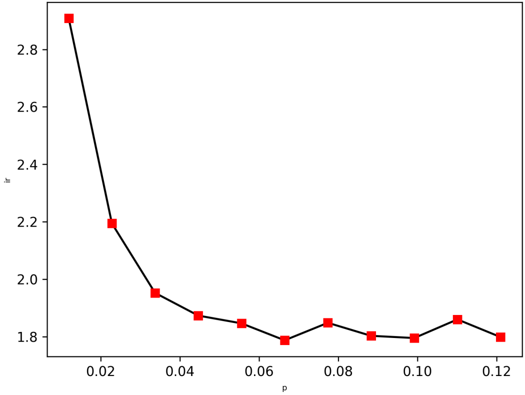
Interest rate

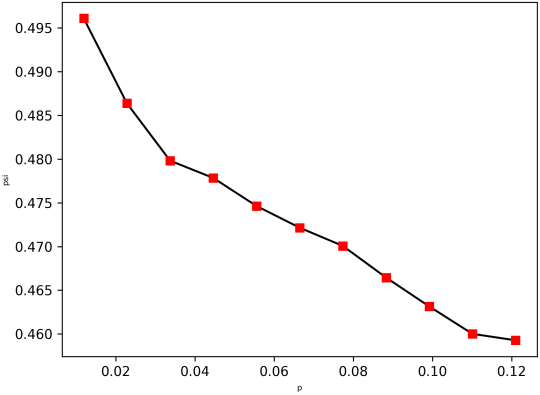
As the probability of attachment p increases, the interest rate declines. This reflects a more competitive market: borrowers can choose among more lenders, reducing lenders' market power. Lower rates benefit borrowers but also compress lender margins and increase exposure to default.



p	r^i
0.012	2.9083
0.023	2.1940
0.034	1.9521
0.045	1.8731
0.056	1.8461
0.066	1.7871
0.077	1.8480
0.088	1.8027
0.099	1.7949
0.110	1.8598
0.121	1.7982

Market power ψ

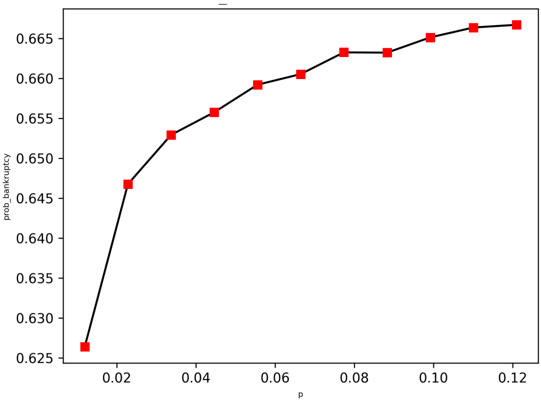
Market power ψ decreases monotonically as connectivity p increases. In a denser network, competition among lenders intensifies, eroding the monopolistic markup. This is the mechanism through which connectivity drives interest rates downward.



p	ψ
0.012	0.4961
0.023	0.4864
0.034	0.4798
0.045	0.4778
0.056	0.4746
0.066	0.4721
0.077	0.4701
0.088	0.4665
0.099	0.4631
0.110	0.4600
0.121	0.4593

Probability of bankruptcy p_b

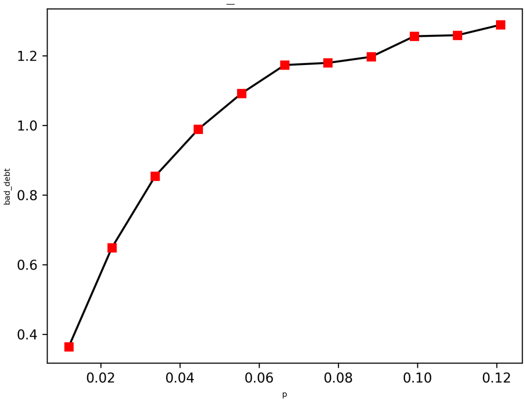
As connectivity increases, the probability of borrower bankruptcy rises. Lenders attempt to serve a larger number of clients, and the decline in interest rates reduces the screening incentive, making the system more fragile.



p	p_b
0.012	0.6264
0.023	0.6468
0.034	0.6529
0.045	0.6558
0.056	0.6592
0.066	0.6606
0.077	0.6633
0.088	0.6632
0.099	0.6652
0.110	0.6664
0.121	0.6667

Bad debt

Bad debt grows with connectivity: more loans are extended as p increases, and a higher volume of lending translates into greater accumulated losses from default. This amplifies the systemic cost of low interest rate environments.



p	B
0.012	0.3636
0.023	0.6486
0.034	0.8536
0.045	0.9888
0.056	1.0917
0.066	1.1733
0.077	1.1794
0.088	1.1968
0.099	1.2559
0.110	1.2590
0.121	1.2888

Outline

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Key Findings

As connectivity p increases:

1. The Giant Component Size increases sharply above the percolation threshold.
2. Market power decreases, and the equilibrium interest rate falls.
3. The probability of borrower bankruptcy rises, as lenders serve a larger number of clients.
4. Bad debt accumulates, reflecting greater systemic exposure to default.
5. Taken together, these results show that **an interest rate that remains too low for too long increases both the incidence of bankruptcies and the spread of contagion.**

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Key Findings

1. **Failure composition shifts**: rationing dominates at low p; repayment failure dominates at high p as interest rates decline.
2. The two opposing channels—lower rates from competition, and higher bankruptcy risk from greater connectivity—imply a **non-trivial trade-off** for regulators.
3. Excessively low interest rates, while beneficial for borrowers in the short run, can be **detrimental to the stability of the system** in the long run.

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Policy Implications

- **Heterogeneity** is a root cause of systemic instability; the failure of large banks has disproportionate systemic effects.
- Interbank lending should ideally be restricted among banks with **similar liquidity characteristics.**
- Reserve requirements dampen cascades but **constrain growth.**
- **Monetary policy implications:** an interest rate environment that remains too low for too long can increase connectivity-driven fragility, suggesting that macroprudential policy must account for the endogenous response of network structure.